

Make Your Own GSM Security Switch To Prevent Vehicle Theft

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This simple GSM security switch project is based on Arduino Nano microcontroller (MCU) and SIM800L GSM modem (Fig. 1).

The cellular-based alert system is configured to send an SMS in response to a security switch action. When the security sensor switch input recognises an event, the system sends a customised alert message to a predefined cellphone number.

Circuit and working

The circuit diagram of the GSM security switch is shown in Fig. 2. It is built around Arduino Nano board (Board1), SIM800L GSM modem (Board2), 5V, 2A power supply and a few other components.

Arduino Nano runs on 5V DC supply, and the modem gets around 4.3V DC. When hardware setup is ready, upload the sketch to Arduino Nano board. Power up Arduino Nano and GSM, and take a trial run. Do not forget to insert a valid micro-SIM card in the slot provided in GSM modem before powering up the system. Nor-

mally, the SIM card registers automatically to the home network within a couple of seconds. Onboard LED indicator of SIM800L modem will blink once every two to three seconds (not continuously) when it successfully registers to the home network.

A micro-USB cable is used to connect Arduino Nano with the PC or laptop for uploading the source code.

After uploading GSMsecurity.ino to Arduino, remove the micro-

USB cable and connect 5V, 2A power supply to the circuit. LED1 is the power-on indicator and LED2 is the system status indicator.

Resistors R1 and R2 are current limiters, while the rest (R3 through R5) form a tricky logic-level translator setup. Add diode D1 and buffer capacitor C1 to Vcc line of SIM800L module.

Connecting pin D9 of Arduino to ground will trigger an alert in the cellphone. That is, if a low signal is received through a sensor connected across connector CON2, it will trigger the alert generator instantly. You can use this circuit as a security alert system, such as for vehicle anti-theft alert system, burglar alarm and so on.

Software

The software is written in Arduino programming language. GSMsecurity.ino sketch is compiled and uploaded to Arduino Nano board using Arduino IDE.

There are many featured Arduino libraries that you can try (the one from Cristian Steib is steibkhriz@gmail.com), but the sketch included here will work well for this application without such libraries.

Here, communication between Arduino Nano and SIM800L goes over SoftwareSerial library and uses standard AT commands. So, simply use this sketch after changing the recipient's

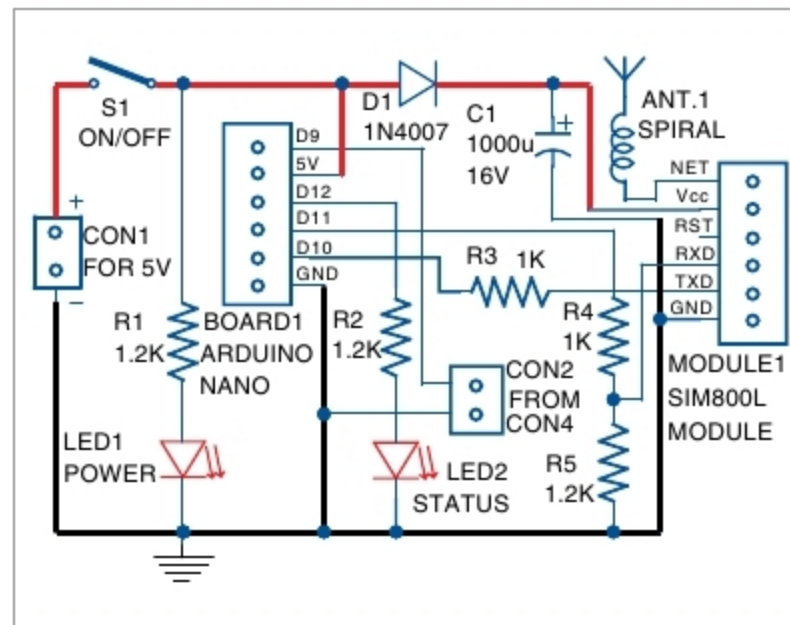


Fig. 2: Circuit diagram of the GSM security switch

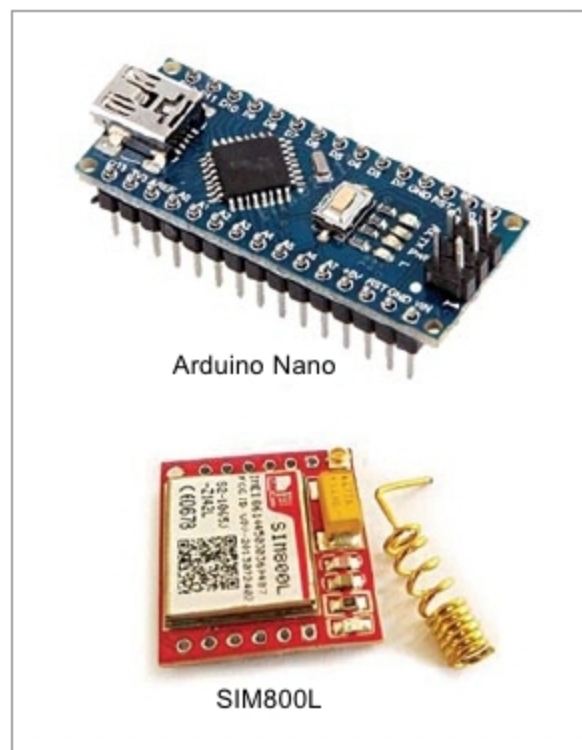


Fig. 1: Arduino Nano board and SIM800L modem

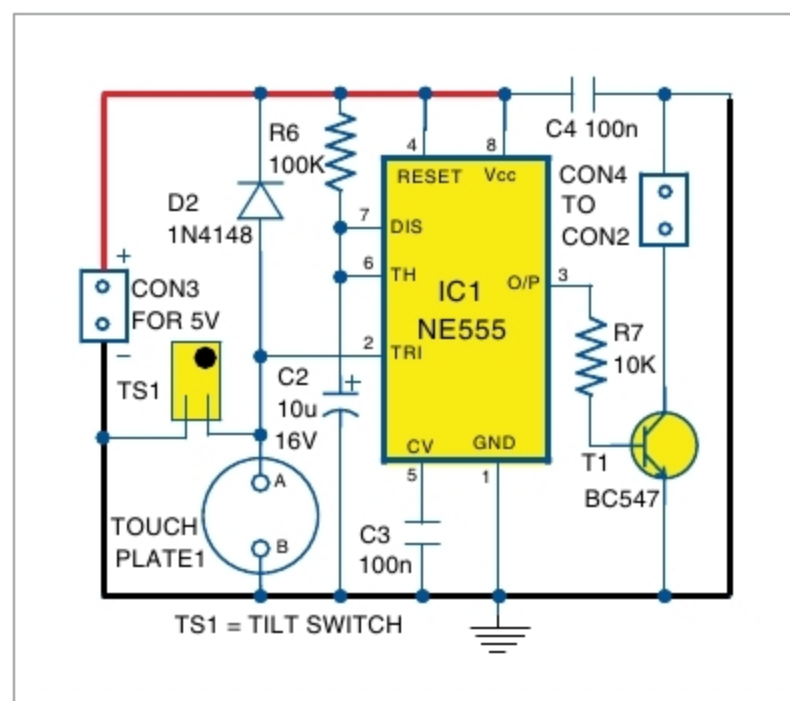


Fig. 3: Circuit diagram of the touch/tilt sensor for vehicle anti-theft alert